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Identity Development During the College Years: Findings From the West Point Longitudinal Study

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Using R. Kegan's (1982, 1994) theory of identity development, it was discovered that most military cadets enter college with relatively simple ways of making meaning, and for many the college years are a time of significant developmental change. For most, the key developmental issue is not self- authorship but establishing shared meaning.

To the extent that human development is a principal aim of higher education (Chickering & Reisser, 1993), pursuit of that aim is facilitated by theories that illuminate the nature of psychological development during the college years. Although there is no shortage of such theories (Creamer, 1990; Pascarella & Terenzini, 1991), most theories relevant to college student development focus on one domain of experience. These include the domains of intellectual understanding (King & Kitchener, 1994; Perry, 1970), moral and ethical reasoning (Kohlberg, 1976; Rest, 1986), identity (Baxter Magolda, 2001; Erikson, 1968), and ego development (Loevinger, 1976). In the current investigation we explored Robert Kegan's theory of meaning-making, where he identified the progressive shifts in perspective-taking that may underpin and set the stage for changes in other developmental domains (Kegan, 1982, 1994). Kegan argued that one's perspective-taking capacities illuminate a broad range of psychological phenomena, including motivation, interpersonal relations, sense of self, the emergence of idealism, metacognition, perceptions of fairness, and the ability to avoid what has been termed adolescent impulsivity.

In the current study we hope to provide educators and college student development professionals with a framework for better understanding the developmental achievements and constraints of college-age students by examining both longitudinally and cross-sectionally the Kegan developmental levels of cadets attending a selective four-year institution of higher education, the United States Military Academy. In addition to providing a characterization of the developmental levels of first- and second-year college-age students, we examine the extent to which students show developmental changes in meaning-making perspectives from the first to the senior year. Our findings are consistent with findings of significant developmental change during the college years (Baxter Magolda, 1992; King & Kitchener, 1994) and suggest that the key

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developmental issue for most students is establishing shared meaning, what Kegan (1994) terms "co-construction." Further, what both Kegan (1982, 1994) and Baxter Magolda (2001) term "self-authorship" begins later in the college years, and appears to be a central developmental issue for a relatively few students. We will argue that in seeking to help students become self-authoring it is important not to lose sight of the earlier developmental transformation, from self-interest to shared meaning, because this transition is the key developmental issue for the majority of our students.

Robert Kegan's Constructive-Developmental Theory of the Self

In the tradition of Piaget (Inhelder & Piaget, 1958) and subsequent neo-Piagetian theorists (Kohlberg, 1976; Perry 1970), Kegan (1982, 1994) posits progressive qualitative shifts in how people construct and understand the world around them, a process that both defines and influences identity, self-concept, and interpersonal relationships. The focus of Kegan's theory is broader than both Piaget's theory of progressively more complex ways of knowing the physical world and Lawrence Kohlberg's theory of progressively more complex processes of moral decision making. Like both of these theorists, Kegan postulates that each successive developmental stage builds upon and subsumes earlier developmental stages. As such, later or "higher" stages are characterized by the development of more complex and more encompassing ways of structuring experience. Later stages are not achieved until the cognitive structures of earlier stages have been fully developed.

Successive stages are constructed in response to environmental complexity and challenge that render one's current stage too narrow and simplistic for effective adaptation. In the absence of such complexity or challenge, development is not expected to occur. Kegan describes five developmental stages beyond infancy, three of which are relevant to development during the college years: Stages 2, 3, and 4.

Kegan's Stage 2: The basis of preadolescent experience. At Stage 2, individuals demonstrate a capacity to organize specific events and experiences into concrete conceptual categories and to view events from multiple perspectives. As a result, Stage 2 individuals are able to take others' perspectives (see things from another person's perspective as distinct from one's own) and understand that others often have points of view and agendas different from one's own. But even though they can see things from another's perspective (something children at Kegan's Stage 1 cannot do), Stage 2 individuals tend to view others' perspectives and interests in terms of the implications of those perspectives for the Stage 2 individual's *own* perspective and interests. As a result, Stage 2 individuals tend to define their interpersonal relationships in terms of reciprocal exchange (i.e., competing interests) and fairness. So, for example, if you are perceived as someone who controls important resources (e.g., you are a parent, teacher, dating partner, or residence hall counselor) the Stage 2 individual often approaches you with the implicit question, "What can I do for you so you'll be inclined to help me meet, or at least not interfere with, my goals?" If, instead, you are perceived as having no such resource power, the Stage 2 individual will often react to you in an egocentric, dismissive fashion. Kegan suggests that because Stage 2 individuals look at the world through the organizing lens of their self-interests, their self-esteem is derived principally from competent performance of specific valued role behaviors (e.g., attending class, wearing the right clothes, attending church)

and achievement of concrete, external goals (e.g., attracting responsive dating partners, getting good grades, getting invited to join a sorority or fraternity).

According to Kegan (1982), Stage 2 perspective-taking begins to emerge around age 6 or 7 in most individuals and is thought to be fully developed by about age 12. Consistent with his assertion that developmental progression is not inevitable but depends on level of environmental challenge and support, Kegan presents empirical evidence that some individuals never progress beyond Stage 2 (Kegan, 1994, pp. 194-195). Kegan's findings are consistent with recent research using Loevinger's measure of ego development level (Westenberg & Gjerde, 1999). These researchers found that although 14-year-olds tended to be highly similar in their level of ego development (Levels E3 through E4), there was considerably more variability in developmental level by the time these same individuals reached age 23 (Levels E3 through E8). So even if those of us involved with college students expect most college students to develop beyond Kegan's Stage 2, we should expect at least some to remain at Stage 2 throughout the college years.

Kegan's Stage 3: Moving into a world of shared meaning. At Kegan's next developmental stage (Stage 3), individuals demonstrate a new capacity to organize the concrete categories of Stage 2 thinking into abstract, hypothetical categories and to view experiences from multiple perspectives simultaneously. The capacity for simultaneous perspective taking enables Stage 3 individuals to be identified with, rather than merely interested in, interpersonal and societal expectations. For the first time at Stage 3, the individual's sense of self (one perspective) is in part a function of how he or she thinks others view him or her (a second perspective), and these two perspectives are taken simultaneously (how I experience myself being experienced by others). Self-esteem moves from a basis in competent performance and goal accomplishment at Stage 2 to a basis in the internal sense of being well regarded by others at Stage 3.

Simultaneous perspective taking also permits one to be self-reflective in a new and more "psychologically minded" fashion. For the first time, one's internal experience is chiefly a function of how one experiences and thinks about one's inner qualities and emotions (i.e., feelings about one's feelings and thoughts about one's thoughts). In contrast, because of their centeredness in competent performance and goal achievement, Stage 2 individuals' internal experience is a function of the success they are having in meeting their own goals and agendas. If they are successful, they feel good, and can report this experience to others. If they are unsuccessful, they feel bad or wronged and can tell others as much.

The transition between Stages 2 and 3 tends to occur over a period of months or even years. During this transition, individuals alternate between the simpler and more concrete perspectives of Stage 2 and the more complex, encompassing and abstract perspectives of Stage 3, gradually coming to rely more fully on the latter for attaching meaning to their experiences. Kegan conveys the impression that this transition from Stage 2 to Stage 3 typically takes place in adolescence stating, "It makes sense to conceive of the period between 12 and 20 as a time during which normal mental development consists in the gradual transformation of mind from the second to the third (stage)" (1994, p. 37). Given that the transition typically takes a year or two, Kegan's statement suggests that most individuals begin *and complete* the transition out of Stage 2 during the teen years. The

current study bears directly on this prediction.

Kegan's Stage 4: Achieving self-authorship. At Kegan's Stage 4, individuals reconstruct and psychologically decenter from their Stage 3 social and interpersonal identifications using individually constructed values and standards. Whereas Stage 3 individuals define themselves in terms of co-constructed interpersonal and social expectations (because they are embedded in simultaneous perspective-taking), Stage 4 individuals construct personal values and standards and then define themselves in terms of how well they are meeting those self-authored standards. The resulting independence from (ability to take a perspective on) social and interpersonal expectations enables Stage 4 individuals to make decisions and commitments in the absence of shared social support and in the face of competing social and organizational expectations. Kegan argues that most high-level managerial and leadership positions require a Stage 4 conceptual capacity for success (Kegan, 1994, pp. 137-186). If this is true, then progressing from Stage 2 through Stage 3 to Stage 4 is critical to the growth of autonomous professionals, professionals of the sort capable of exercising sound judgment in the face of the complex, ambiguous, and rapidly changing situations that increasingly characterize modern work life.

Very few college-age individuals (those in the 17 to 22 year range) begin developing beyond Stage 3 functioning toward Kegan's Stage 4. In short, progression to Stage 4 is probably *not* a typical feature of psychological development in one's early 20s. Indeed, studies reported by Kegan (1994, p. 195) suggest that Stage 3, not Stage 4, is the modal stage of adulthood. Furthermore, it appears that most adults never develop to Kegan's Stage 4, at least not before late middle age, the oldest age cohort systematically studied in the research summarized by Kegan (1994, pp. 187-197) and Lewis and Jacobs (1992). Nonetheless, the current brief description of Kegan's Stage 4 is useful to clarify the limitations of Stage 3 and because Kegan's Stage 4 includes some attributes that are often identified among the broader goals of a college education, such as independence of thought and personal authority (Baxter Magolda, 2001; Forsythe, Snook, Lewis, & Bartone, 2002). If it turns out that very few undergraduate students are in the progression from Kegan's Stage 3 to Stage 4, these educational goals may be largely unattainable during the college years.

The Current Study

The current paper is a summary of the first comprehensive investigation of Kegan's developmental theory in traditional-age college students and describes the developmental levels of military cadets both cross-sectionally and longitudinally. As such, the paper provides preliminary answers to basic theoretical questions about the developmental level of college-age students as well as whether the perspective-taking of students becomes progressively more complex across the college years. The findings contribute to a growing literature on the cognitive capacities of college students (Baxter Magolda, 1992; King & Kitchener, 1994; Pascarella & Terenzini, 1991) and provide educators with a view of both the fundamental capacities and the fundamental limitations students bring to the classroom as well as to their personal lives.

METHOD

Overview

Subject-object interviews, developed by Kegan and his colleagues to assess individuals' meaning-making stage (Lahey, Souvaine, Kegan, Goodman, & Felix, 1988), were

conducted on two random samples of United States Military Academy (USMA) cadets who were members of the class of 1998. The initial sample included 38 students who were interviewed early in the Fall of their first year of enrollment. Thirty-one were reinterviewed during the Spring of their second year. In anticipation of natural attrition from the academy (typically close to 22%) and subsequent nonparticipation by some study participants, a second random sample of 24 additional participants was interviewed during the Spring of their second year. From the combined second year sample of 55 participants, 35 were reinterviewed during December of their fourth year at the academy. Of these, 21 had been interviewed during their first year, 34 during their second year, and 20 at all three points. All interviews were audio recorded, transcribed, and scored by trained scorers. The interview scores were examined cross-sectionally by class year and longitudinally between the second and fourth year to provide a characterization of student developmental levels at three class years as well as an estimate of the prevalence of changes in developmental level during the college years.: To assess the predictive validity of subject-object interview scores, a performance measure that is a standard part of students' military education was obtained from students' files. Finally, an opportunity sample of 20 first-semester students attending a large state university was also assessed using the Kegan subject/object interview to gauge the extent to which military academy students are developmentally similar to students attending a selective, nonmilitary university.

The Subject-Object Interview

All participants were interviewed in accordance with the standard interview protocol established by Kegan and his colleagues at Harvard's Graduate School of Education (Lahey et al., 1988). Using this procedure, a trained interviewer prompts respondents to recall their own recent experiences in response to a stimulus word or short phrase printed on a series of 10 index cards. The stimulus words, reflecting emotion-laden themes, include terms such as "sad," "success," "torn," "anxious and nervous," and "important to me." After being instructed to recall experiences, the respondents are instructed to begin the interview proper by describing any one of the recalled events. The interviewer then asks a series of follow-up questions intended to elicit the underlying perspective the respondent is using to make sense of the event being described. When the interviewer is able to elicit the respondent's most encompassing meaning-making perspective, the interviewer invites the respondent to describe another experience that came to mind in response to the same or another card. Respondents typically talk about four to seven experiences during an assessment interview.

When there is adequate interviewer inquiry and participant response, developmental level scores can be assigned to what are termed "scorable bits." Scorable bits may be as short as a single sentence or as long as several paragraphs. Only responses that reflect the participant's underlying meaning-making structures are scorable. The remainder of a participant's responses is considered unscorable content. Scores may be assigned for responses that clearly reflect achievement of a full stage (i.e., Stage 2, 3, or 4) or as falling at one of four distinguishable points in the transition between two adjacent full stages. In the current study these fine grained scoring distinctions are not reported, and students are shown as either having achieved a full stage (Stage 2 or Stage 3) or as being in one of three transitions, Stage 1 to Stage 2 (only one student was found

to be in this transition), Stage 2 to Stage 3, or Stage 3 to Stage 4.

First- and second-year interviews were conducted by any two members of a four-person team of trained interviewers. These included three officer/psychologists in the Department of Behavioral Sciences and Leadership at the USMA and a fourth research psychologist employed by the U.S. Army Research Institute for the Social and Behavioral Sciences. Scoring of the first-year interviews was done independently by each of the four members on the four-person team. Where there were disagreements, scores were discussed and a consensus score assigned. The current first-year developmental level scores reported were the consensus scores. Average interrater agreement across the four scorers for the 38 first-year interviews was 63% (to within 1/5 stage) and 83% (to within 2/5 stage).

Sophomore and senior year cadet interviews were conducted either by a member of the four-person team described above (GBF, the second author of the current paper) or another trained rater (PL, the first author of the current paper). PL also conducted the 20 comparison interviews of freshmen attending a large state university. The intermixed sophomore and senior year interviews were scored by either GBF or PL over a period of several months more than 9 months after the senior year interviews had been completed. The scorers were blind to both the identity of the participants and to the school year (sophomore or senior) during which each interview was conducted. Interrater agreement among the various scorers was established as follows: PL and GBF each independently scored six interviews that had been conducted several years earlier at the Command and General Staff College of the U.S. Army. With respect to the scoring distinctions used in the current research, there was 100% interrater agreement for these six interviews. PL and GBF also both scored 9 of the 20 freshman interviews conducted at the nonmilitary state university. There was complete agreement (stage or transition between two stages) for seven of the nine common interviews (78%). The remaining two interviews were scored in adjacent scoring categories by the two scorers. Because GBF's individual scores of the first-year military academy sample correlated .87 with the first year consensus scores used in the present study, it is presumed that there was an acceptable level of correspondence between the Time 1 consensus scores and the sophomore and senior scores.

Criterion Measure

To address the question of whether developmental level makes any difference in how students actually function, a routinely collected measure of cadet performance was accessed from file data, the Military Development (MD) grade, which reflects overall performance in fulfilling military training requirements. The MD grade is a weighted average of performance ratings by several supervisors (U.S. Corps of Cadets, 1995; Schwager & Evans, 1996). A military officer supervisor provides half of the MD grade, and the other half is an average of grades given by 2 or 3 upper-class cadet supervisors. Students receive an MD grade, similar to a course grade, at the end of each academic semester. For the current investigation, average MD scores were computed for the first 2 years and for the final 2 years. Military training for first- and second-year cadets is oriented primarily toward mastering assigned military tasks in the role of subordinate, such as carrying out tasks assigned by higher rank cadets. In contrast, military training for upper-class cadets focuses on planning, organizing, and performing as cadet leaders in the organization.

Participants

All military cadet interviews were conducted at the USMA at West Point, New York. The academy is a highly selective, racially and geographically diverse four-year military college that provides students with both a traditional academic education as well as extensive military training. Unlike most colleges, only 16% of USMA students are female, and entering students cannot be younger than 17 or older than 23 years of age nor can they be married. The average age of the cadets included at the inception of the current study was 18.6 years. Five of the 38 cadets interviewed during the first year (13%) were female. Seven of the 52 cadets interviewed during the sophomore year (13%) were female. None of the 32 cadets who returned for senior-year interviews were female. These cadets were randomly selected from among members of the Class of 1998, which was being followed as part of a large investigation of cadet leader development. USMA cadets are admitted on the basis of outstanding academic, athletic, and leadership performance during high school. See Bartone, Snook, and Tremble (2002) for a more complete description of the larger Class of 1998 study.

To determine the extent to which the developmental level scores of the military cadet sample were comparable to those of non-military college students, 21 freshmen enrolled in a large, selective state university were also assessed for developmental level. These 21 were chosen from among 132 freshmen who completed a demographic survey for extra credit points in their introductory psychology class. On the demographic survey, students were asked to provide their age, highest SAT and ACT college aptitude test scores, high school GPA, and to list important high school extracurricular activities, including any leadership positions held during their high school years. Students were also asked to indicate whether they had prior military service (one of the 21 did), their interest in serving in the military following graduation, and their willingness to participate in a subsequent 60 to 90 minute assessment interview. In an attempt to obtain a comparison sample that resembled the USMA cadet sample, only those university students with high SAT or ACT scores (as reported by the student), and involvement in varsity athletics and/or high school leadership positions were invited to be interviewed. Recording equipment malfunctioned during one university student interview (a 23-year-old student with prior military service), so that subject's interview could not be scored. Twenty scorable interviews were obtained. The mean SAT score for the first-year cadets was somewhat higher than that of the university students, 1210 versus 1122, but the mean ACT scores were identical for the two samples (27.6). The mean age of the two samples was virtually identical (18.6 versus 18.5 years). The major demographic difference between the two samples was a lower proportion of females in the cadet sample than in the university sample (13% versus 35%).

RESULTS

Two basic questions were of particular interest in the current research: (a) what is the Kegan developmental level of traditional college-age cadets, and (b) to what extent are the college years a time of significant developmental change for military cadets?

Comparison of Stage Scores Across Years

Table 1 shows the stage scores for military college students at Time 1 (early first year), Time 2 (late sophomore year), and Time 3

TABLE 1.

Percentage of Kegan Stage Scores at Each of 3 Year Levels for West Point Cadets

	Time 1 Freshman Year % (*n*)	Time 2[a] Sophomore Year % (*n*)	Time 3[a] Senior Year % (*n*)
Stage 2	21.0 (8)	23.1 (12)	6.3 (2)
Stage 2 to Stage 3 Transition[b]	63.2 (24)	52.0 (27)	31.3 (10)
Stage 3	15.8 (6)	19.2 (10)	43.7 (14)
Stage 3 to Stage 4 Transition[c]	0.0 (0)	5.7 (3)	18.7 (6)
	n = 38	*n* = 52	*n* = 32

[a] Three interviews each during the sophomore and senior years were unscorable.

[b] The Stage 2 to Stage 3 transition includes the four scorable distinctions between these two full stage scores (Lahey et al., 1988).

[c] The Stage 3 to Stage 4 transition includes the first three of four scorable distinctions between these two full stage scores (Lahey et al., 1988).

(midsenior year) for all participants who had scorable interviews. Different numbers participated at each of the three times, because 24 new participants were added at Time 2 and not all participants participated in follow-up interviews. Scores are grouped into four categories: Stage 2, Stage 3, those in transition between Stage 2 and Stage 3, and those in transition between Stage 3 and Stage 4. No participants were found to be functioning at a full Stage 4. Only a small percentage of the first year cadets (15.8%) had achieved Kegan Stage 3 (see Table 1). The cadets in their sophomore year (Time 2) scored slightly higher than the first-year cadets (Time 1). More striking are increases in the percent of participants scored as functioning at Kegan's Stage 3 and above in the senior year (Time 3). By the end of the sophomore year only 25% were at or above Stage 3. By the senior year, this percentage had more than doubled to 62.5%. These findings suggest that most traditional-age college student cadets begin college at Kegan's Stage 2 or in the transition to Kegan's Stage 3. By the senior year most, but not all, have reached a full Kegan Stage 3, and a small number may be starting the transition from Stage 3 to Stage 4.

Individual Change Scores

A more direct assessment of developmental change can be gained by examining the Kegan stage scores of the 29 participants (all males) who had scorable interviews during both their sophomore and senior years. Eighteen of the 29 (62%) showed developmental progression. Only 2 participants showed regression. Of the 9 participants who showed no change from their sophomore to their senior year, 4 were already at Stage 3 during their sophomore year, a stage beyond which most college students are not expected to move until somewhat later in adulthood, if at all (Kegan, 1994). These longitudinal data provide direct support for the proposition that for many undergraduate male cadets the college years are a time of significant developmental change.

From a theoretical perspective, the two instances of regression are unexpected, because once the capacity to construct a broader and

more encompassing perspective of a higher stage is attained, it should not be possible to lose that perspective. One can, of course, elect to construct meaning in a given situation employing a simpler meaning-making framework than one is capable of using, but when conducting a subject/object interview (Lahey et al., 1988) interviewers are expected to press interviewees to express their highest level of functioning. It is worth noting that in the longitudinal investigation of developmental levels reported by Kegan (1994, pp. 189-190) there were also several instances of apparent developmental regression.

Correlates of Developmental Level Scores

Substantial variability in participants' developmental levels is evident in the scores reported above. If identity status really matters, as Kegan and other developmental theorists suggest, then variations in developmental level should be correlated with other indicators of adaptation or functioning. At the USMA all cadets are evaluated with respect to their performance as cadet subordinates and as cadet leaders through the assignment of military development grades (see above). We examined the relationships among cadets' Kegan developmental level scores at each of the three class years and these same cadets' combined upper-class and combined lower-class MD grades. Most striking was a Spearman's rho correlation of .41 ($p < .01$, $df = 29$) between senior-year subject-object interview scores and each senior's composite junior-senior MD grade. This correlation indicates that senior cadets who are determined to be more psychologically mature on the basis of a Kegan subject/object interview were rated by tactical officers and cadet leaders (a combined rating) across their junior and senior year as having more effectively carried out their cadet leader responsibilities than were senior cadets who were determined to be less psychologically mature. In contrast, the correlation between sophomore year subject/object interview scores and each cadet's sophomore composite MD grade during the first 2 years was statistically indistinguishable from zero (Spearman's rho = .04, $df = 51$). It should be noted that it is only during the junior and senior years that cadets have meaningful leadership responsibilities. Thus, it appears that with respect to rated performance of cadet upper-class leadership duties, but not performance of

TABLE 2.

A Comparison of Frequency of Developmental Level Scores Between Nonmilitary First-Year University and Cadet Students and As (Percentages) of Each Sample

Developmental Level[a]	University Students *n* (%)	Military Cadets *n* (%)
Stage 1 to Stage 2 Transition[b]	1 (5%)	0 (0%)
Stage 2	6 (30%)	8 (21%)
Stage 2 to Stage 3 Transition	8 (40%)	24 (63%)
Stage 3	5 (25%)	6 (16%)
	n = 20	*n* = 38

[a] Assessed from interview transcripts in accordance with Lahey et al. (1988) scoring guide.

[b] The student included here was found to be functioning at the last scorable distinction prior to a full Stage 2 score.

more routine lower class cadet responsibilities (e.g., appearance of the cadet uniform, neatness of one's barracks room), perspective-taking level appears to matter, and higher developmental levels are associated with higher rated performance.

The State University Comparison Sample

Table 2 shows the data comparing subject/object interview scores of USMA first-year cadets with a similar selective sample of first-year students attending a large state university. As can be seen, the percentages at each of the stage levels are similar. The two score distributions were subjected to an independent samples *t* test analysis. Stage scores were coded so that the lowest score obtained (the highest scorable distinction in the transition between Stage 1 and Stage 2) was assigned a value of 5 and each successively higher score, including the four scorable distinctions between each full stage, were assigned successively higher integers. In this system, a Stage 2 score was coded as a 6 and a Stage 3 score as an 11. The mean stage score for the twenty university students was 7.95, and the mean stage score for the 38 cadets was 8.24. This difference did not reach statistical significance, $t(56) = -0.57$, $p = .57$ (two-tailed).

Qualitative Analyses

The qualitative differences in meaning-making between students functioning at Stage 2 and those functioning at Stage 3 can be appreciated better by examining several interview excerpts. Consider the following excerpts from an assessment interview with a sophomore cadet who recalls feeling angry that two of his friends from his first year were kicked out of school for lying. Although he describes being angry at a third cadet who got his two friends into trouble, he also says he is angry at his two friends. Notice how he constructs this anger.

> I think the worst part was I didn't . . . not want to see them every day and not be together and stuff. I never had the friends that I have until I came here. And I remember telling them, "I can't believe you guys are leaving here." . . . So I was angry at them. Because we went through a lot together and we had so many good times for the year that we knew each other. It was just great then.

His anger doesn't appear to relate to any abstract principle here, instead he is angry because he has been deprived of something he valued, friends he could have fun with. Consistent with Kegan's description of Stage 2 individuals, he is looking at this situation through the organizing lens of his own interests. His interests are thwarted, so he reacts with anger. He is aware that his two friends have their own interests when he later states "But, you know, my friends, they still wish they were back, back with us." But seeing their loss and regret appears to have little or no moderating effect on his anger toward them. In the language of Kegan's theory, he is able to see two different perspectives (a Stage 2 capacity) but he is unable to make another's perspective a part of his own, a capacity that emerges for the first time at Stage 3.

Contrast the above cadet's anger with another sophomore cadet's anger who was assessed as functioning at Kegan's Stage 3. This cadet is angry because his Tactical Officer insisted that a cadet platoon stand out in the rain while undergoing an inspection of their adherence to haircut standards. Significantly, this cadet was not personally involved. He didn't have to stand in the rain. He was only an observer.

> It made me angry that he made those 36 men stand outside in the rain. I wondered if that was something that would really

> happen in the army. . . . And so it made me really angry inside that he (the officer) talked about taking care of your soldiers and stuff. And then I see him out there in the rain going through this inspection. . . . As far as good leadership and stuff, that was kind of questionable at least. . . . You know it was a kind of "practice what you preach" kind of thing. He was kind of going against what he had seemed to talk to us about all the time.

This student's anger is generated by a more complex (Stage 3) construction of this incident than was evident with respect to the Stage 2 student's anger. What is at issue here is not a thwarted personal agenda or interest. Instead this student is embedded in a generalized expectation (a good leader takes care of his or her subordinates) and its violation by a member of the profession (an officer). What seems to make him angry is a violation of a shared expectation of good leadership, an expectation that he uses to define his sense of self. His anger entails simultaneous perspective taking, where his sense of self (one perspective) includes another perspective (the generalized expectation that members of one's profession should display good leadership). His reaction results from the same sort of Stage 3 capacity that is necessary for the expression of adolescent idealism and the transcendence of concrete self-interest.

The qualitative differences between Stage 2 and Stage 3 students can also be seen in the different ways they construct their understanding of leader-subordinate relationships. Consider the following interview excerpt from another cadet sophomore. He describes why he feels more motivated when he has a cadet platoon sergeant whom, as he states it, "puts out 110%."

> The leaders that I've had in the past, the ones that are really good, I perform well in there. If they are putting out 110%, and they are doing their best, you can visually see it; their uniform looks really cool. You know, if their shoes are really shined, you know, mine should get that good too. And if I see them putting 110% out for us, they're doing it for us—their own troops—and they really care about them, then I'm going to work that much harder.

What sounds Stage 2 about this excerpt is that the cadet constructs the relationship between himself and his platoon sergeant as a sort of exchange—he works hard at being a good cadet, so I'll work hard at it too. This view can be accommodated by the Stage 2 capacity to take two perspectives, the cadet's and the platoon sergeant's, one at a time. Still, it is possible that this cadet's orientation to the platoon sergeant's "caring" has become an internal feature of the cadet's own experience of himself. This would require a Stage 3 capacity to take two perspectives and hold them together simultaneously. The interviewer seeks to clarify this issue with this same cadet (the interviewer's questions are shown in all capitals).

> WHY IS IT IMPORTANT TO YOU THAT THEY CARE?
>
> I guess for me its because of the tangible rewards of what will happen if somebody does really care about you. Maybe, like for instance, if they are putting 110% out, then you're going to put 110 % out there. And then you are all going to feel better about everything else.
>
> BECAUSE YOU THINK YOU'LL BE ABLE, YOURSELF, TO SUCCEED MORE?
>
> Yeah, yeah, definitely.

So we see that the interviewer's questions lead to the notion that the cadet will do more and thereby get more of his own goals met, if he has a motivated leader (the "tangible

rewards"). And presumably he "feels better about everything else" because he, individually, is being more successful. This, and other portions of the interview, lead to the conclusion that this cadet had not yet begun to move past a Stage 2 way of constructing his relationships.

We can contrast the above excerpt with the description of leadership found in the interview of a sophomore cadet scored at Stage 3.

> I think the best cadet is the one who really understands that his real mission in life—in the Army—is being there for his troops. I mean he's got to have a commitment to the Army's goals overall.
>
> HOW DO YOU KNOW THAT'S THE RIGHT THING?
>
> I guess its one of those gut feelings that you have. And I'm of the firm belief that if you don't want that, if you're here just for yourself, you know, get the heck out of here. And, you know, the Academy kind of makes you realize that taking care of your troops is the most important thing.
>
> SO YOU HAVE THAT RESPONSIBILITY. WHERE DOES THAT RESPONSIBILITY COME FROM?
>
> I think the responsibility as a leader to take care of your troops comes from the fact that you're the lieutenant. In addition to the fact that you're a human being. You know, as a human being you have a responsibility, I think, to take care of everybody else.

What makes this a Stage 3 response is this cadet's embeddedness within another point of view *as the source of* his own point of view. Those appropriated points of view include the academy's expectations of an army officer, the responsibilities of having a professional position ("you're the lieutenant"), and an idealistic societal view of the responsibilities of membership in the human race. Missing is an embeddedness in his individual needs. Indeed, he critiques self-interest when he admonishes those who are "here just for yourself" to "get the heck out."

Many of the cadets interviewed during their first and second years for the current investigation were found to be in the transition between Kegan's Stage 2 and his Stage 3. The following excerpts from a university sample participant provides a flavor of that developmental movement, the experience of beginning to make sense of the world in a new and more inclusive manner. The student describes an experience of going out to lunch with his mother on a recent trip home from college.

> I went to lunch with my mom on Saturday and it was really nice. Because that was the first time I really *wanted* to go to lunch with her, and that surprised me right off the bat.
>
> WHAT WAS YOUR MOTIVATION FOR WANTING TO GO WITH HER?
>
> Just the friendship; because I cared for her. It was always there, but I'd always taken it for granted. But now I'm realizing that she's going to be a very big part of my life from now on. And what made it so special was that I was the one that wanted to go to lunch, and I set it up. Before, she'd beg me to go to lunch and I never wanted to do it. And now I was just excited that it was me—I had the feelings to want to do it.

What is still Stage 2 about this excerpt is his centeredness in his own needs and desires. He didn't used to want to hang out with his mom; now he does. But there is more than simple self-interest evident here. Most striking is a new level of self-reflectiveness. It isn't merely that he has different interests; instead, he is struck by how his interests have shifted.

His focus is internal. He is having feelings about his feelings. It also appears that the nature of his feelings has shifted. Before, he and his mother had competing interests—she wanted to go to lunch; he didn't. Now he wants to go. Why? "Because I care for her." In a manner that is not seen in someone functioning at Stage 2, there seems to be no instrumental purpose behind this new feeling. His agenda here is merely to give expression to an internal feeling and share his feelings with another. This orientation to shared experiences expresses the Stage 3 capacity to hold two experiences together simultaneously. The interviewer wants to make sure this is what is happening.

> WHY WAS IT IMPORTANT FOR HER TO KNOW ABOUT WHAT YOU WERE DOING AND WHAT YOU WERE EXPERIENCING?
>
> Because I knew she would care about it. And having someone who cares about what you have to say is a big deal. At least I feel it is. You'd rather talk to someone who really cares deeply about what you're saying. So the fact that I knew she cared a lot and that she was honest and sincere made it much better.

That he has not completed the Stage 2 to Stage 3 transition is evident in the fact that his focus is still primarily on his own experience. And it is also the case that his mother's "caring deeply" seems to remain more an external event than a part of his inner experience of himself. Simultaneous perspective taking, a hallmark of Kegan's Stage 3, is not yet fully present in this excerpt.

DISCUSSION

The current findings suggest that traditional-age military college students in their first, and even in their second year in school, are not as psychologically mature as Kegan had originally assumed. Across the first- and second-year samples of the current investigation, less than 1 in 5 military academy cadets were found to be functioning fully at Kegan's Stage 3. And even though 50% to 60% of these cadets were transitioning to this stage, almost all of these students were unable to completely transcend concrete thinking and self-interest (hallmarks of Kegan's Stage 2). Furthermore, fewer than 1 in 4 equally select first-year students from a nonmilitary university comparison sample were found to be functioning at a full Kegan Stage 3.

By the senior year, more than a third of our cadet participants had still not completed the transition to Stage 3. In contrast, another 19% had begun the transition to Kegan's Stage 4. Though somewhat surprising, the modest level of mental complexity found in many participants in the current study is consistent with findings from other recent studies of the cognitive-developmental capacities of traditional-age college students (Baxter Magolda, 1992; King & Kitchener, 1994). Baxter Magolda, studying "ways of knowing" in 101 traditional-age Miami University undergraduates, found only one student who was using "independent knowing" by the sophomore year. By the senior year, only 16% of Baxter Magolda's participants were classified as independent knowers, and only 2% were using "contextual knowing," the highest of four progressively more complex "domains" in her epistemological reflection model.

The current findings indicate that many traditional-age military college students and perhaps nonmilitary college students as well, understand their *early* college experience in terms of concrete rewards and personal achievements. Much of the time they don't think abstractly; their relationships with others are based on exchange rather than mutuality;

they are not psychologically minded or introspective; and they have difficulty sustaining a focus on their own long-term interests. Although most of these students are working hard to succeed and to meet social expectations for good behavior, they do not begin college able to make the welfare of others or of their college community a part of their own *internal* experience. Lacking the capacity for simultaneous perspective taking that is the hallmark of Kegan's Stage 3, these developmentally "younger" students instead view their college community primarily as a venue for acquiring and expressing academic and social competence.

Arguably, these are not the sort of students most of us expect to encounter in our work as educators and college student development professionals. Chronologically at the very end of their teen years, we expect the majority of our students to think and act more like adults than like junior high school students, so we are surprised and not a little disappointed when they don't. Most of us were attracted to higher education settings out of an eagerness to help students grow into thoughtful critical thinkers, young adults whom we can assist on the path to self-authorship and psychological autonomy. But the message of the research reported here is that many of our students are fully engaged in another, developmentally earlier, transition. Before they are capable of becoming self-authoring independent thinkers, many of our students will first need assistance in thinking collectively, relativizing self-interest, and psychologically joining communities of shared learning and shared meaning-making. This Stage 2 to Stage 3 transition is qualitatively different than the development of self-authorship (Kegan's Stage 3 to Stage 4 transition), and facilitating this earlier transition requires particular sorts of programs and strategies.

There are, without doubt, many successful programs and classroom strategies that provide the effective mix of support and challenge required for facilitating Kegan's Stage 2 to Stage 3 developmental transition. Consider, for example, the role of rules of conduct, campus regulations, honor codes, and other aspects of what can be termed environmental structure. Although the past several decades have seen a retreat from codes of conduct on many college campuses, it is also clear that some colleges have begun to return to formulating clear expectations for student conduct, expectations that they then enforce with specified consequences. Though we are not accustomed to thinking about rules, regulations, and codes of conduct in this way, they may in fact constitute an essential "support" for the further development of Stage 2 students.

But, by themselves, clear and consistently enforced codes of conduct do not facilitate the Stage 2 to Stage 3 transition, because they provide insufficient challenge to Stage 2 meaning-making. What is needed are policies and programs that disrupt Stage 2 students' views of codes of conduct as solely a feature of the external environment. The goal is to replace compliance with internalization and self-regulation, where students become a part of the standard-generating community and the community's standards become a part of the way students regard themselves.

Even though Kegan's Stage 2 to Stage 3 developmental transition is about a kind of internal self-regulation, it is not about self-authorship. Stimulated by the gradual acquisition of Stage 3's simultaneous perspective-taking, the Stage 2 to Stage 3 transition's internalization and self-regulation are, in actuality, *shared* activities (i.e., "This is the way *we* should act"). This developmentally earlier sort of "self-regulation" flows from experi-

encing oneself through the eyes of the community. Facilitating this shift to simultaneous perspective-taking entails designing highly structured codes of conduct that include processes that lead to the "productive undoing" of Stage 2 self-interest. This can be accomplished in a number of ways, such as the employment of group decision making and consensus, creating task interdependence, and by insuring that the community includes Stage 3 students and staff members who are available to welcome the transforming student into the college community. The goal is for Stage 2 students to join the rule-constructing community, to subordinate self-interest to the interests of the community, so that the success of the group is experienced as one's own, and failures are assessed not in terms of their personal cost but in terms of the cost to the community (Kegan, 1994, p. 47).

One such program that appears to promote this progression at the USMA is the Cadet Honor System. At first the honor code is seen by many cadets as an externally imposed set of rules of conduct and associated penalties for failure to comply. But as cadets become increasingly involved in running the honor system and struggling to make sense of its impact on themselves and their classmates, most come to see its underlying logic and find it becoming a part of their self-identity. What was powerfully external because of its relevance to Stage 2 students' self-interest, gradually takes up residence in the emerging Stage 3 students' capacity to make the community's welfare a part of their own sense of self. Baxter Magolda (1992, 2001) has provided a number of similar accounts of features of the college environment that promote this powerful transition.

But what about those students who arrive at college already functioning at Kegan's Stage 3 or who complete the Stage 2 to Stage 3 transition relatively early in their college years? These are the students who properly need an environment that self-consciously seeks to promote independent thought and self-authorship. The challenge for those of us who work with diverse college populations is creating classroom and extracurricular programs that not only help complete that Stage 2 to Stage 3 transition for many students but also promote the Stage 3 to Stage 4 transition in that important minority that are ready to begin the path to self-authorship. In our cadet sample, 47% of the seniors interviewed had completed the Stage 2 to Stage 3 transition and another 19% were in transition to Kegan's self-authoring stage. It is with this group of students that Baxter Magolda's recommendations for facilitating self-authorship (Baxter Magolda, 2001) should find fertile ground. From a Kegan developmental perspective, what appears to be necessary are learning and living contexts that expose students to multiple contexts where students' various shared identities (fraternity/sorority member, loyal friend, future spouse, religious person, etc.) come into conflict with one another. Programs that get students off campus, such as study abroad, community service projects, work-study experiences, and interdisciplinary research and demonstration projects, may work well, particularly when they provide opportunities for students to reflect upon and stake out those elements that will gradually become the basis of an evolving self-authored identity.

Limitations and Future Directions

The two major limitations of the current investigation both concern the generalizability of the findings. Although based on a carefully constructed random sampling procedure, military college cadets' development may not be typical of the identity development of other

types of college students. The current participants met stringent admission requirements with respect to academic, athletic, and leadership accomplishments. In addition, the participant sample was nearly all male. Inclusion of a similarly accomplished sample of first year students at a nonmilitary university led to the suggestion that the developmental level of new military college students may not be much different than the developmental level of similar nonmilitary first-year college students. Nonetheless, until further research is undertaken with other college student populations using Kegan's theory as the guiding developmental framework, it is not possible to conclude that the current findings are typical of the general college student population.

The second major limitation of the current findings concerns the impact of the highly structured military college environment on students' development. The USMA differs from the typical college environment in a number of respects. To mention just a few of the differences, all students are in their late teens or early 20s, all live on campus in barracks where they are subject to a set of rules and regulations that constrain behaviors from room cleanliness to when lights have to be extinguished. All cadets are required to participate in military training, be members of sports teams (intercollegiate, club, or intramural), and to attend class and other campus events. Considerable effort is expended in attempting to foster "character development" through all aspects of the 47-month cadet experience (United States Military Academy, 2002). Few colleges and universities have this level of structure or this level of total institutional commitment to the development of the whole student. To the extent that the cadet environment actually impacts student psychological development, then one should be cautious in generalizing the current findings concerning developmental change across the college years to college students in other contexts.

Developmental research is costly and highly demanding. Assessment of students' developmental levels requires time-consuming individual interviews, and reliably scoring those interviews demands extensive training. Prospects for objectively scored developmental assessments are mixed, in part due to the reality that assessing something as complex as how individuals construct meaning requires an assessment approach that can capture that complexity. The work is difficult, but the eventual payoffs are well worth the effort in terms of a gradual increase in understanding college student development and how it can be promoted.

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